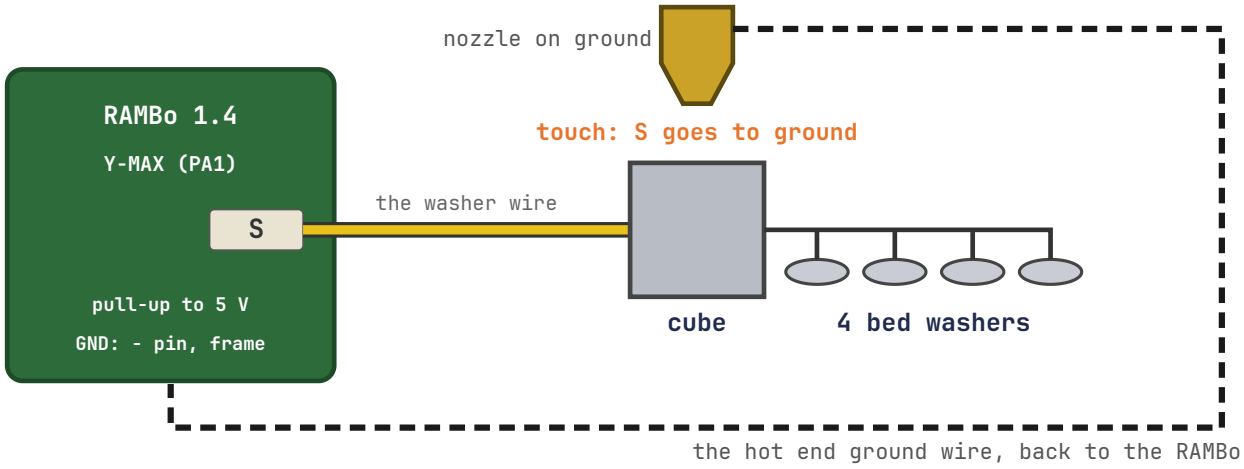
LULZBOT TAZ WORKHORSE · RAMBO 1.4 · KLIPPER

The cube wire on Y-MAX

The steel cube at the front right of the bed measures the backlash of X, Y and Z. The nozzle touches it, as G425 did in the LulzBot firmware. Its wire was on Z-MIN, and the BLTouch has Z-MIN now. The wire goes to the free Y-MAX header.



How it works

- The cube and the four bed washers are on one wire, the washer wire. The RAMBo holds the wire at 5 V.
- The nozzle is on ground. When it touches the cube, the wire goes to ground, and the deck sees a trigger.
- `CUBE_MEASURE` touches the top, the right, the left and the back of the cube. After each touch the nozzle backs away in 0.01 mm steps. The distance to the open contact is the play of the axis.

What you need

- The washer plug. It came out of Z-MIN when the BLTouch cable went in.
- A multimeter with a continuity beep.
- A brass brush for the nozzle tip.
- If the plug is lost: a Molex SL 3-pin housing (50-57-9403) and one SL socket (16-02-0086), as on the other endstop plugs.

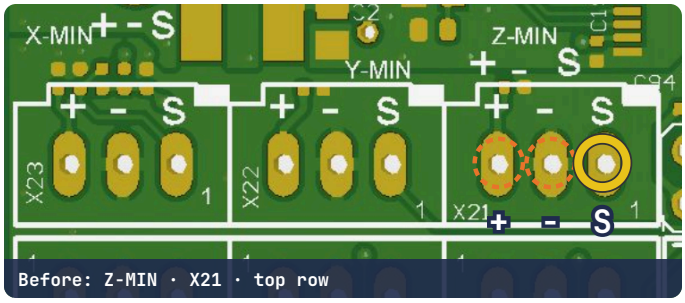
Why

The LulzBot firmware had backlash correction: `M425 X0.25 Y0.25 Z0.04`. Klipper has none. The measure shows if the play is still there after the new belts. Fix play in the belts, the pulleys and the idlers first.

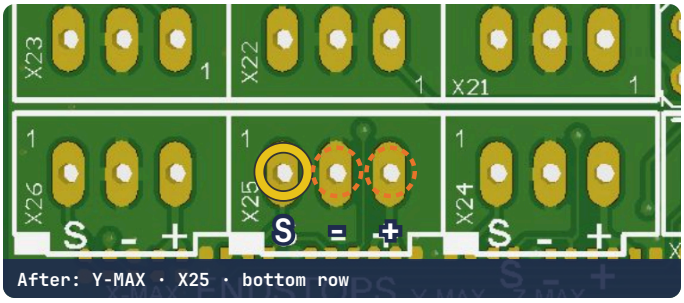
Step 1: check the wire

1. **Turn off the printer** and unplug it. Open the control box.
2. **Find the washer plug.** It hangs near the ENDSTOPS block of the RAMBo, with one wire on the S position.
3. **Cube.** Put one probe of the meter in the S socket of the plug, and the other on the cube. The meter must beep. If it does not, the cube has its own wire: follow it from the cube, and tell Claude where it goes.
4. **Washers.** Do the same with a bed washer. The meter must beep.
5. **Nozzle.** Put one probe on the nozzle, and the other on the − pin of any endstop header. The meter must beep. Clean the nozzle tip with the brass brush first: plastic on the tip stops the contact.

Step 2: the plug on Y-MAX



Before: Z-MIN · X21 · top row



After: Y-MAX · X25 · bottom row

- ENDSTOPS has two rows of three headers. The top row is X-MIN, Y-MIN, Z-MIN, with **+ − S** from the left. The BLTouch is on Z-MIN.
- The bottom row is X-MAX, **Y-MAX**, Z-MAX, with **S − +** from the left. The right Z switch is on X-MAX and the left one on Z-MAX.
- The headers of the bottom row turn the other way. Put the latch of the plug on the same side of the header as on Z-MIN. Then the wire stays on **S**.

1. **Push the washer plug onto Y-MAX**, the middle header of the bottom row, with the wire on S (the left pin).
2. **Check** with the meter: S of Y-MAX to the cube beeps. S of Y-MAX to the + pin does not beep.
3. **Close the control box.**

Picture: the top of the RAMBo 1.4a, from the UltiMachine design files (github.com/ultimachine/RAMBo-1.4).

Step 3: the config and the test

```
[printdeck_cube]
pin: ^!PA1      # Y-MAX
left: 261       # X of the left face
right: 271      # X of the right face
back: -14       # Y of the back face
top: 3          # Z of the top face
```

The section

The deck update of the backlash PR has the section in `printer.cfg`. The places come from the LulzBot firmware. They are estimates: `CUBE_MEASURE` reports the places that it finds.

1. **Turn on the printer.** Leave the nozzle cold.
2. In the Console of the deck, send `CUBE_QUERY`. It must say **cube: open**.
3. **Touch the cube to the nozzle** with a short wire: one end on the cube, the other end on the cold nozzle. Send `CUBE_QUERY` again. It must say **cube: TRIGGERED**.
4. If the two answers are the other way round, remove the `!` from the pin, and test again.

Step 4: measure

Clean nozzle, BLTouch pin up

A plastic film on the tip stops the contact, and the nozzle then presses into the cube. Each touch stops 1 mm past the place in the config (0.5 mm on the top), so a failed contact cannot press farther.

1. **Clean the nozzle tip** with the brass brush, warm (150 °C for PLA). Then wait for 100 °C or less.
2. **Home:** `G28`. Check that the BLTouch pin is up.
3. **Measure:** `CUBE_MEASURE`. The nozzle goes to the cube at the front right, touches the top, then the right, the left and the back faces. It takes a few minutes.
4. **Read the result** in the Console, for example:
   `Backlash: X 0.240 mm (right 0.250, left 0.230), Y 0.180 mm, Z 0.040 mm`

What the numbers mean

Up to about 0.05 mm is good. More than 0.1 mm on X or Y shows play: check the belt tension, the set screws of the motor pulleys, and the idlers, and measure again. Tell Claude the result. The deck can then correct the rest of the play, as M425 did in the LulzBot firmware.